

## CREATING A SPRING BOOT APPLICATION

### pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="https://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.4</version>
    <relativePath/>
  </parent>

  <groupId>com.library</groupId>
  <artifactId>LibraryManagement_Boot</artifactId>
  <version>1.0-SNAPSHOT</version>

  <name>LibraryManagement_Boot</name>

  <properties>
    <java.version>21</java.version>
  </properties>

  <dependencies>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
```

```

        <artifactId>spring-boot-starter-data-jpa</artifactId>
    </dependency>

    <dependency>
        <groupId>com.h2database</groupId>
        <artifactId>h2</artifactId>
        <scope>runtime</scope>
    </dependency>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
    </dependency>

</dependencies>

<build>
    <plugins>

        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>

    </plugins>
</build>

</project>

```

### **application.properties (under resources):**

```
spring.application.name=LibraryManagement_Boot
```

```

spring.datasource.url=jdbc:h2:mem:librarydb
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=

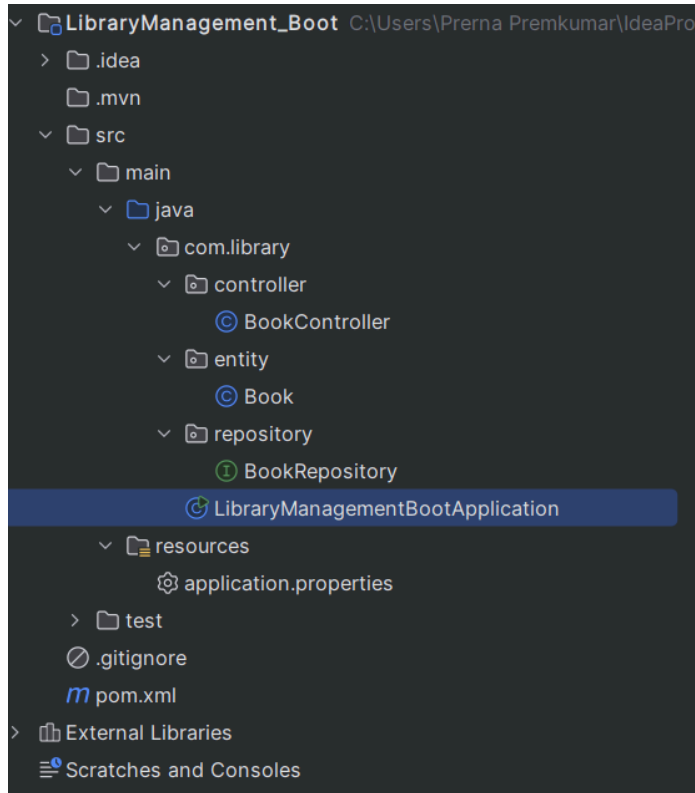
```

```
spring.jpa.hibernate.ddl-auto=update
```

spring.jpa.show-sql=true

spring.h2.console.enabled=true

## Project Structure:



## BookController.java:

```
package com.library.controller;
```

```
import com.library.entity.Book;
import com.library.repository.BookRepository;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;
```

```
import java.util.List;
```

```
@RestController
@RequestMapping("/books")
public class BookController {
```

```

@Autowired
private BookRepository repository;

@GetMapping
public List<Book> getBooks() {
    return repository.findAll();
}

@PostMapping
public Book addBook(@RequestBody Book book) {
    return repository.save(book);
}
}

```

### **Book.java:**

```

package com.library.entity;

import jakarta.persistence.Entity;
import jakarta.persistence.Id;

@Entity
public class Book {

    @Id
    private Long id;

    private String title;

    private String author;

    public Book() {
    }

    public Book(Long id, String title, String author) {
        this.id = id;
        this.title = title;
        this.author = author;
    }
}

```

```
public Long getId() {  
    return id;  
}  
  
public void setId(Long id) {  
    this.id = id;  
}  
  
public String getTitle() {  
    return title;  
}  
  
public void setTitle(String title) {  
    this.title = title;  
}  
  
public String getAuthor() {  
    return author;  
}  
  
public void setAuthor(String author) {  
    this.author = author;  
}  
}
```

#### **BookRepository.java:**

```
package com.library.repository;  
  
import com.library.entity.Book;  
import org.springframework.data.jpa.repository.JpaRepository;  
  
public interface BookRepository extends JpaRepository<Book, Long> {  
  
}
```

#### **LibraryManagementBootApplication.java:**

```
package com.library;
```

```

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class LibraryManagementBootApplication {
    public static void main(String[] args) {
        SpringApplication.run(LibraryManagementBootApplication.class, args);
    }
}

```

The screenshot shows an IDE with the following components:

- Project Explorer:** Shows a project named 'LibraryManagement\_Boot' with a package structure including 'entity' (containing 'Book'), 'repository' (containing 'BookRepository'), and 'resources' (containing 'application.properties'). The 'LibraryManagementBootApplication' is also visible.
- Editor:** Displays the 'LibraryManagementBootApplication.java' file with the following code:
 

```

import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class LibraryManagementBootApplication {
    public static void main(String[] args) {
        SpringApplication.run(LibraryManagementBootApplication.class, args);
    }
}

```
- Run Console:** Shows the output of the application. The first part of the output is:
 

```

main] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache disabled
main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection conn0: url=jdbc:h2:mem:librarydb user=SA
main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
'ool-1')

```

 The second part of the output is:
 

```

main] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA
main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
main] JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed d
main] o.s.b.a.h2.H2ConsoleAutoConfiguration : H2 console available at '/h2-console'. Database available at 'jdbc:h2:mem:librarydb'
main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
main] c.l.LibraryManagementBootApplication : Started LibraryManagementBootApplication in 7.45 seconds (process running for 8.197)

```

### Sample application:

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Prerna Premkumar>curl -X POST http://localhost:8080/books -H
"Content-Type: application/json" -d "{\"id\":1067,\"title\":\"Java Pro
gramming\",\"author\":\"Prerna P\"}"
{"id":1067,"title":"Java Programming","author":"Prerna P"}
C:\Users\Prerna Premkumar>
```

